

PROJECT REPORT

SMART AA BATTERY CHARGER WITH AUTOMATIC CUT-OFF

1. Introduction

This project is about building a smart battery charger for rechargeable AA Ni-MH batteries. Standard cheap chargers do not stop charging when the battery is full, which can overheat the battery and ruin its life.

This smart charger fixes that problem. It uses a special IC to send a safe, steady current to the battery, and another IC to monitor the voltage. The moment the battery is fully charged, the circuit automatically stops charging and switches the indicator lights to let the user know.

2. Project Features

- **Safe Charging:** It charges the batteries with a safe, low current so they do not get hot.
- **Auto Cut-off:** It automatically stops charging the exact moment the battery is full.
- **Clear LED Indicators:** It uses three distinct lights to show the exact status:

I. Red LED Only: Power is turned ON, but no battery is inside.

II. Red + Yellow LEDs: The battery is currently charging.

III. Red + Green LEDs: The battery is completely full.

3. Component List (Parts Used)

Component Name	Description / Specification	Quantity
LM317 IC	Adjustable voltage regulator used to	1

	maintain a steady charging current	
TL431 IC	Precision voltage sensor that acts as the main automatic cut-off switch	1
BC547 Transistor	NPN electronic switch used to turn the Green LED ON when full	1
BC177 Transistor	PNP electronic switch used to turn the Yellow LED OFF when full	1
1N4007 Diode	Silicon rectifier diode that blocks backward battery current drain	1
5.6 Ohm Resistor	1W power resistor used to set the constant current to 223mA	1
1k Ohm Resistors	1/4W resistors used to protect the LEDs and transistor bases	4
10k Ohm Preset	Adjustable trimming potentiometer used to set the full-voltage limit	1

Current Control:

The LM317 IC and the 5.6 Ohm resistor work together to create a steady current of around 223mA. This current flows safely into the battery through the 1N4007 diode, which acts like a one-way valve.

Voltage Sensing:

The 10k Preset continuously takes a small sample of the battery's voltage and feeds it into the TL431 IC.

- While Charging: The battery voltage is low. The TL431 stays turned OFF. This leaves the BC177 transistor ON, lighting up the Yellow LED (Charging).
- When Fully Charged: The battery reaches its full voltage limit. The TL431 detects this and instantly turns ON. It drops its output to Ground, which turns the BC177 OFF (Yellow LED goes out) and turns the BC547 ON (Green LED lights up).

5. Circuit Assembly and Setup

6. Soldering: Place all the components on the Veroboard according to the circuit diagram and solder them carefully. Double-check the pins of the BC547 and BC177 transistors so they do not get swapped.
7. Testing Power: Plug in the 12V DC adapter without putting any batteries in the holder. The Red LED should turn ON.
8. Calibration (Setting the Limit): Put a voltmeter across the battery holder terminals. Slowly turn the 10k Preset screw until the Green LED just pops ON.
 - Set it to 2.85V if you are using a 2-slot battery holder.
 - Set it to 5.7V if you upgrade to a 4-slot battery holder.

9. Charging: Put in your empty AA batteries. The Green LED will turn off, and the Yellow LED will turn on, showing that charging has started.

10. Conclusion:

This project successfully demonstrates how a few basic analog components can be used to make a smart automatic battery charger. It does not require any programming or expensive microcontrollers. The circuit is highly reliable, cheap to build, protects the batteries from overcharging, and gives the user clear visual updates using simple indicator lights.